

Automated Usability Evaluation in Virtual Reality: A Systematic Mapping Study - Supplementary Material

DENNIS MÜHLENSTÄDT, ENES YIGITBAS, and STEFAN SAUER

We used specific search queries for the digital libraries, but with common logic:

```
Title OR Keywords OR Abstract
should match
Virtual Reality AND usability AND evaluation
```

Including combinations of OR connected synonyms for *Virtual Reality*, *usability*, and *evaluation*.

1 Springer Link

```
(
  "Virtual Reality" OR "VR"
  OR "Augmented Reality" OR "AR"
  OR "Mixed Reality" OR "XR"
  OR "Extended Reality"
)
AND (
  "usability" OR "user experience" OR "UX"
)
AND (
  "evaluating" OR "testing" OR "evaluation"
  OR "test" OR "validation" OR "validating"
)
```

2 Scopus

```
TITLE-ABS-KEY(
  (
    "Virtual Reality" OR "VR"
    OR "Augmented Reality" OR "AR"
    OR "Mixed Reality" OR "XR"
    OR "Extended Reality"
  )
  AND (
    "usability" OR "user experience" OR "UX"
  )
  AND (
    "automated" OR "automation" OR "automatic"
    OR "agent based" OR "agent-based"
  )
  AND (
    "evaluating" OR "testing" OR "evaluation"
    OR "test" OR "validation" OR "validating"
  )
)
```

```

)
AND(
    LIMIT-TO ( DOCTYPE , "cp" )
    OR LIMIT-TO ( DOCTYPE , "ar" )
)
)

3 IEEE Xplore
(
    (
        "Document Title":"Virtual Reality"
        OR "Document Title":"VR"
        OR "Document Title":"Augmented Reality"
        OR "Document Title":"AR"
        OR "Document Title":"Mixed Reality"
        OR "Document Title":"XR"
        OR "Document Title":"Extended Reality"
    )
    AND (
        "Document Title":"usability"
        OR "Document Title":"user experience"
        OR "Document Title":"UX"
    )
    AND (
        "Document Title":"evaluating"
        OR "Document Title":"testing"
        OR "Document Title":"evaluation"
        OR "Document Title":"test"
        OR "Document Title":"validation"
        OR "Document Title":"validating"
    )
)
OR (
    (
        "Author Keywords":"Virtual Reality"
        OR "Author Keywords":"VR"
        OR "Author Keywords":"Augmented Reality"
        OR "Author Keywords":"AR"
        OR "Author Keywords":"Mixed Reality"
        OR "Author Keywords":"XR"
        OR "Author Keywords":"Extended Reality"
    )
    AND (
        "Author Keywords":"usability"
        OR "Author Keywords":"user experience"
    )
)

```

```

    OR "Author Keywords": "UX"
  )
  AND (
    "Author Keywords": "evaluating"
    OR "Author Keywords": "testing"
    OR "Author Keywords": "evaluation"
    OR "Author Keywords": "test"
    OR "Author Keywords": "validation"
    OR "Author Keywords": "validating"
  )
)
OR (
  (
    "Abstract": "Virtual Reality"
    OR "Abstract": "VR"
    OR "Abstract": "Augmented Reality"
    OR "Abstract": "AR"
    OR "Abstract": "Mixed Reality"
    OR "Abstract": "XR"
    OR "Abstract": "Extended Reality"
  )
  AND (
    "Abstract": "usability"
    OR "Abstract": "user experience"
    OR "Abstract": "UX"
  )
  AND (
    "Abstract": "automated"
    OR "Abstract": "automation"
    OR "Abstract": "automatic"
    OR "Abstract": "agent based"
    OR "Abstract": "agent-based"
  )
  AND (
    "Abstract": "evaluating"
    OR "Abstract": "testing"
    OR "Abstract": "evaluation"
    OR "Abstract": "test"
    OR "Abstract": "validation"
    OR "Abstract": "validating"
  )
)
)

```

4 The ACM Guide to Computing Literature

Title:(

```

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    OR "test" OR "validation" OR "validating"
  )
)
)
OR Abstract:(
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  AND (
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  AND (
    "automated" OR "automation" OR "automatic"
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  AND (
    "evaluating" OR "testing" OR "evaluation"
    OR "test" OR "validation" OR "validating")
)
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```

Table 1: Overview of all included publications and mapping to method, measurement, and evaluated usability dimension

Ref	Methods	Usability Dimensions	Measurements
[1]	automated measurement	physical effort	pressure load
[3]	automated measurement	flow	heart rate, EMG, respiration
[4]	automated measurement	usability, physical effort	hand movement, head movement
[8]	automated measurement	attention, cognitive load	EEG
[10]	automated measurement	physical effort	heart rate
[11]	automated measurement	UX	heart rate, events
[13]	automated measurement, visualization	UX	movement
[14]	automated measurement	presence	heart rate
[15]	automated measurement	presence, area of interest, realism	heart rate, EDA, gaze
[16]	automated measurement	usability, behaviour, engagement	hand movement, head movement, events, gaze
[20]	automated measurement, AutoQUEST	usability smells	events
[21]	automated measurement	engagement	head movement, EEG
[26]	automated measurement	usability	EEG
[28]	automated measurement	learnability	facial expression
[27]	automated measurement	emotions	head movement
[30]	automated measurement, machine learning	anxiety	heart rate
[31]	automated measurement, machine learning	usability	hand movement, head movement, EEG, speech
[32]	automated measurement, AutoQUEST	usability smells	events
[34]	automated measurement	physiological arousal	heart rate, EDA, body temperature
[37]	automated measurement	presence, flow, emotions, cognitive load	EEG, EDA
[38]	automated measurement	stress, ease of use, attention	heart rate, gaze
[40]	automated measurement, machine learning	attention	EEG, pupil
[41]	automated measurement	UX, learnability	decision time
[43]	automated measurement, machine learning	presence	hand movement, head movement, speech, body tracking
[42]	automated measurement, machine learning	presence	hand movement, head movement, speech, body tracking
[44]	automated measurement, visualization	area of interest	gaze
[46]	automated measurement, machine learning	presence	hand movement, head movement, facial expression, scene capture
[47]	automated measurement	attention, UI	head movement, gaze
[48]	automated measurement	area of interest	gaze, body tracking
[50]	agent	UX, learnability, emotions	none
[51]	automated generation, machine learning	realism	none
[52]	automated measurement, visualization	presence, attention, UI, comfort	gaze
[2]	agent	usability, emotions	none
[54]	automated measurement	physiological arousal	heart rate, EDA
[55]	automated measurement	interactability, learnability	head movement
[57]	automated measurement	stress	heart rate
[58]	machine learning	behaviour, sentiment	events, speech
[59]	automated measurement	UX, emotions, stress	heart rate, facial expression, events, speech
[61]	automated measurement	attention, behaviour	head movement, EDA
[62]	automated measurement, visualization	flow, stress	head movement, pupil, movement
[64]	automated measurement	stress, anxiety	heart rate, walking speed
[66]	automated measurement	usability, physical effort	hand movement, head movement, EMG, pressure load